



**Forecasting Housing Density Impacts of Transit-Oriented Development in Leidsche  
Rijn Using a Cellular Automata Land-Use Change Model**

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### Topic and Motivation

The Netherlands is currently suffering from a housing crisis. Within the past ten years, average house prices have increased by almost 200% (CBS, 2026). This bars swaths of people from buying their first home, locking the housing market behind a pay-wall. According to the Dutch central bank, one of the main methods for combating this is constructing new housing (DNB, 2025). This has become one of the main talking points in Dutch politics, especially in political campaigns for the previous election.

During these campaigns, many solutions to the housing problem have been proposed. The election winner D66 introduced the idea of building ten new cities from the ground up. This is a bold and impactful plan, but it raises the question of how these cities are intended to be developed. Their party program states: *“D66 wants to build ten new cities, especially near existing cities, centres or public transport intersections / hubs.”* (D66 Uitvoeringsprogramma 2025-2030).

This strategy is ambitious, and the party does not specify the expected outcomes of the policy. Exact locations or a concrete plan are not given. In addition, the proposal provides limited discussion of the current implementation of transit-oriented development (TOD) in the Netherlands. Existing research suggests that Dutch experts do have an ideal TOD focussed “formula” for developing such new areas, which at their center contain a transit hub. This idea differs from other cultures, especially due to the “extreme” use of bicycles (Pojani, 2015). One of the areas currently being developed which qualifies for TOD is Leidsche Rijn, in the province of Utrecht. Because of its constant state of development in the past years, it provides an excellent candidate for training a model to use for simulation modelling.

### Objective

The aim of this research is to evaluate the transit-oriented development strategy proposed by D66. This is achieved by developing a cellular automata model calibrated on housing data from 2010 and 2015. Using 2017 data, the model will be validated. This validated model

is then used to generate a baseline projection of housing density under development trends for the year 2040. This baseline is compared with an alternative policy scenario that incorporates emphasis on transit points consistent with the D66 proposal. Differences in projected housing between the baseline and policy scenario are used to assess the potential impact of the proposed strategy on urban development in Leidsche Rijn.

### **Literature review**

Existing research suggests that Dutch experts already have an ideal TOD focussed “formula” for developing such new areas, which at their center contain a transit hub (Pojani, 2015). While TOD is not a new concept within city planning and has been actively used in Dutch cities, there are differences between country specific implementations. For example, due to Dutch culture urban areas planned with TOD can be larger, because of the “extreme” use of bicycles. To model this urban growth with some of these specific rules, a few methods can be used.

Nugroho & Al-Sanjay (2018) posited multiple methods for modelling urban growth. Cellular automata, Markov chain, logistic regression, SLEUTH and an artificial neural network were examined and compared. 272 articles were screened for eligibility, with 47 included in the final set. Ultimately, cellular automata models were chosen as the most viable option, because of its strength to be combined with other methods. It was also proven to already be the most utilised method in the screened literature.

To predict land use change and evaluate Transit-Oriented Development (TOD) strategies, researchers in urban simulation have increasingly turned to cellular automata (CA) models. These models are grounded in the theory of emergence (Torrens, 2000), which explains how complex spatial patterns can arise from simple local rules. As Torrens points out, the advantages of using a CA model come from its ability to capture complex spatial rules in a relatively simple manner. Batty et al. (1999) described CA when it was at an early stage in development, using it to model urban development. They also delve deeper into the mathematical workings of CA and its ability to interact with differing cell types such as

housing, road, nature etc. These methods will prove important for this project, forming the basis of our use case.

However, the specific implementation of CA models for forecasting varies significantly across studies, primarily in terms of model calibration methods and the spatial data used. Earlier applications often relied on expert-defined transition rules, which could limit objectivity and replicability. In contrast, a current and robust trend in the literature, as highlighted in the comprehensive review by Gui et al. (2025), advocates for data-driven calibration. Methods such as logistic regression are used to derive transition probabilities from observed historical land-use changes, in order to reduce subjective bias.

At the same time, studies like that of Otgonbayar et al. (2018) provide operational frameworks for defining realistic model states and integrating spatial drivers like distance-decay functions for station accessibility. Despite these methodological advancements, most calibrated CA applications remain oriented toward simulating generic urban growth patterns. Comparatively fewer studies use spatially explicit, calibrated simulation models to generate a baseline forecast against which a specific policy proposal (TOD) can be systematically evaluated. As a result, the outcomes of specific urban plans remain underexplored within the policy analysis literature. Despite advances in data-driven calibration, CA models remain sensitive to parameter choices, highlighting the importance of careful validation and sensitivity analysis.

## **Research Questions**

### ***RQ1***

*How accurately can a cellular automata-based land-use change model, trained on housing development data from 2010 and 2015, reproduce observed housing patterns in Leidsche Rijn in 2017?*

**RQ2**

*To what extent would incorporating D66's transit-oriented development (TOD) strategy into a calibrated cellular automata-based forecasting model alter projected housing in Leidsche Rijn by 2040 compared with a baseline scenario?*

This study is designed to investigate the creation of new cities close to public transport hubs. This is examined by analyzing land-use density datasets from 2010 to 2015, which enables the projection of future urban growth. These projections provide insight into the necessity and relevance of D66's plan to develop cities near major public transport nodes. Using these projections, the difference between projected growth from 2015 onward and growth resulting from the proposed policies can also be examined.

**Datasets**

The first three datasets are from the National Georegister (NGR) of the Netherlands and data was collected by the 'Centraal Bureau voor de Statistiek' (CBS). The land use databases contain the digital geometry of land use in the Netherlands. Examples of land use include traffic areas, buildings, recreational areas, and inland and open water. The boundaries are largely based on the Top10NL (BRT). Aerial photographs are the primary source for interpretation. The data was collected in the summer of 2010, 2015 & 2017. For this study, the land use types for housing and railways were used. Names of the maps: *CBS Land Use 2010 WMS*, *CBS Land Use 2015 WMS*, *CBS Land Use 2017 WMS*.

The fourth and final dataset which was used in this case study is used to get the boundaries of the Leidsche Rijn district. It's also from the CBS and is called "*CBS Wijken en Buurten 2024 WFS*". It is a map made in 2024 which contains all the boundaries of the different districts and neighborhoods in the Netherlands. The links of all these maps can be found in the reference section.

## Method

### Preprocessing

Before addressing the two research questions, some preprocessing was required using QGIS (v3.40.12). First, a polygon layer representing the exact boundary of Leidsche Rijn was created. This was achieved by selecting and exporting the Leidsche Rijn area from the previously mentioned “*CBS Wijken en Buurten 2024 WFS*” dataset.

Next, the CBS land-use maps for the years 2010, 2015, and 2017 were clipped to this Leidsche Rijn polygon, ensuring that only the study area was included in subsequent analyses. Value labels were then assigned to enable model training: residential areas were labeled as “1”, while all other land-use categories were labeled as “0”. The raster size was manually set at 5 x 5 meter, so that the growth could be precisely measured.

Subsequently, a new layer containing the locations of transit points was created. This layer was derived from the 2017 land-use map, with areas classified as “*train area*” labeled as “1” and all other areas labeled as “0”. On top of this layer, a temporary scratch layer was created in which two point features were manually added within the labeled train areas. These points represent the train stations *Utrecht Terwijde* and *Utrecht Leidsche Rijn*.

### RQ1

To answer the first research question: “*How accurately can a cellular automata-based land-use change model, trained on housing development data from 2010 and 2015, reproduce observed housing density patterns in Leidsche Rijn in 2017?*” three steps have been taken. The first step is training a transition probability model using logistic regression on housing data from 2010–2015. The second step is creating a cellular automata model based on the probability map generated by the logistic regression. Finally, the model is calibrated using the observed new builds from 2015–2017. The output of this effort is a calibrated cellular automata land use change model that will be used to serve as the baseline of the model for the second research question.

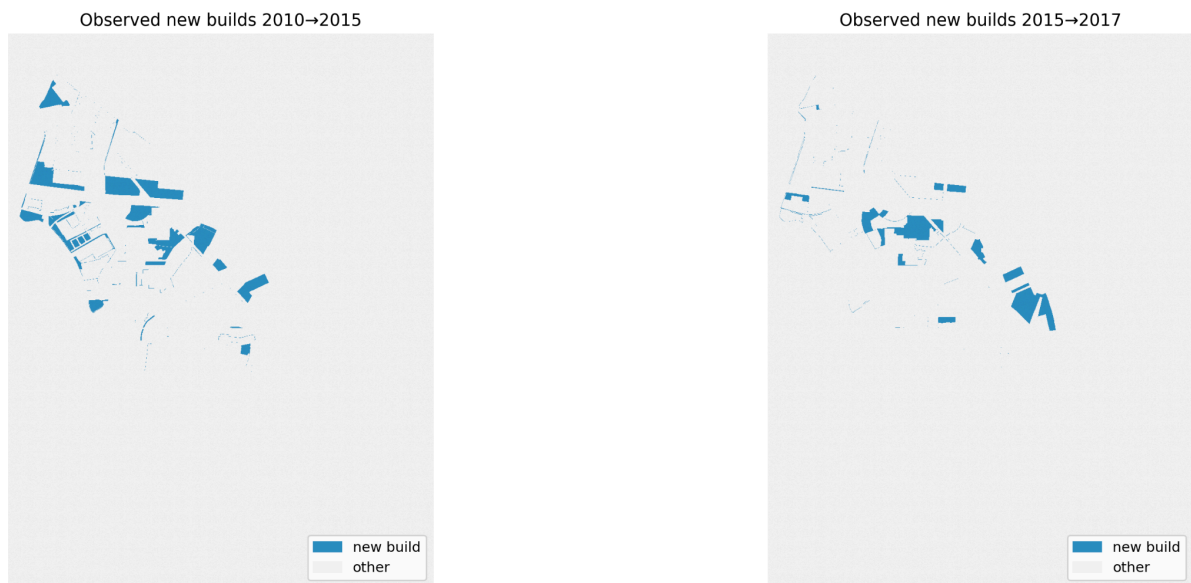
### ***Transition probability model***

For the probability of a cell to become a housing cell a logistic regression model was trained.

In figure 1 the training data is visualised. The map on the left shows all new houses built from 2010-2015.

**Figure 1**

*New housing 2010 - 2015 and 2015 - 2017*



This training data is used to fit the following model:

$$\eta_i = \beta_0 + \beta_1 D_i + \beta_2 N_i(\text{WINDOW\_NEIGH}) + \beta_3 E_i(\text{WINDOW\_EMPTY})$$

Where

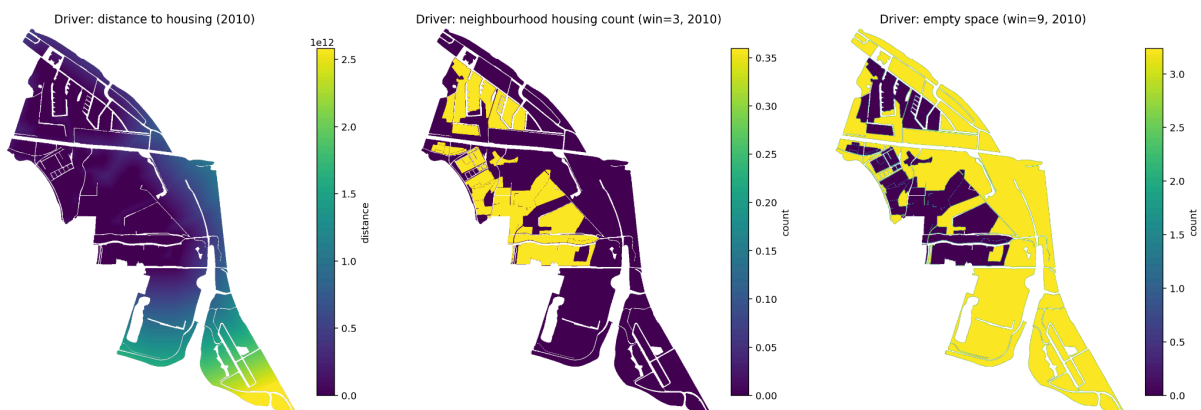
- $D_i$  = distance to nearest existing housing cell
- $N_i$  ("WINDOW\\_NEIGH" ) = number of housing cells within a neighbourhood window of size WINDOW\\_NEIGH
- $E_i$  ("WINDOW\\_EMPTY" ) = number of non-built cells within a neighbourhood window of size WINDOW\\_EMPTY

- $\beta_0$  = intercept
- $\beta_1, \beta_2, \beta_3$  = fitted regression coefficients

This model uses three drivers. The first is the distance to the nearest housing pixel. The second is a grid-based variable that counts the number of housing pixels within each grid cell. The third is also a grid-based variable, but it counts the number of pixels without housing in their surroundings. This final driver was added to capture neighborhoods that are being built in areas where development has not yet occurred. These variables are intentionally included to represent both attraction to existing development and the availability of surrounding undeveloped space. In figure 1 the drivers are visualised.

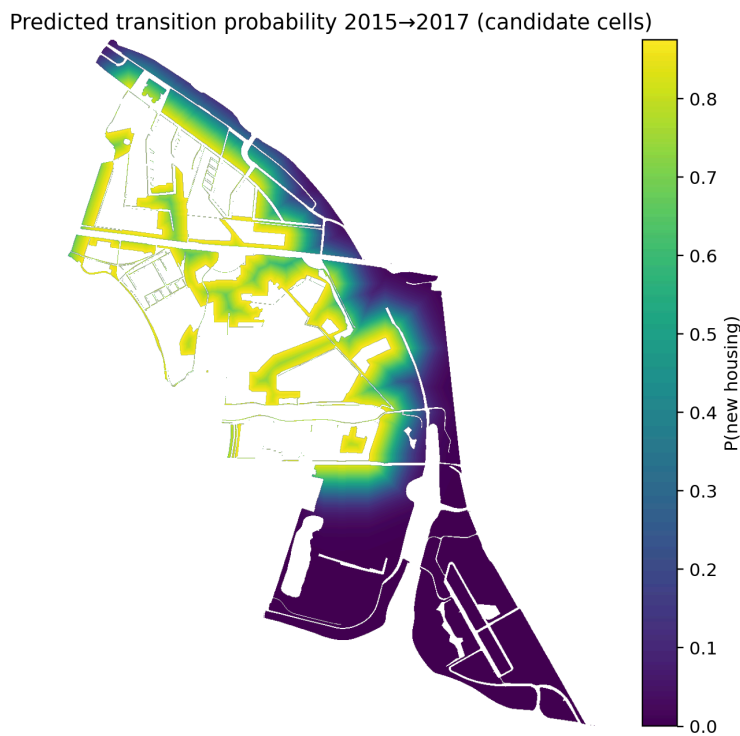
**Figure 1**

*Drivers visualised*



This model outputs a transition probability map that can be seen in figure 2. The map shows the probability for each cell to change from a candidate cell to a housing cell. Yellow signifies a high probability and purple a low probability.



**Figure 2***Transition probability map for housing development in 2015-2017****Calibration process***

All figures that present the model results are based on the calibrated model. The model was calibrated using housing data from 2015–2017. In total, five parameters were calibrated. Two of these are related to the window-based drivers: `window_empty` and `window_neigh`. The remaining three parameters are part of the cellular automata (CA) model: `alpha`, which controls how strongly new housing pixels are attracted to already built pixels in the neighborhood; the `alpha window`, which determines the size of the neighborhood in which `alpha` is applied; and the `seed`, which determines the optimal number of starting points for generating new neighborhoods. The calibrated parameter values are shown in Figure 3.

### Figure 3

*Values used for calibration*

```
WINDOW_EMPTY_GRID = [9, 11, 15, 21, 31]
WINDOW_NEIGH_GRID = [3, 5, 7, 9, 11]

CA_ALPHA_GRID = [0.0, 0.5, 1, 2, 4, 6, 8, 12]
CA_WIN_GRID = [3, 5, 7, 9]
CA_SEED_FRAC_GRID = [0.005, 0.01, 0.02, 0.05]
```

The calibration was based on the Intersection over Union (IoU) of the predicted results. The allocation that performed best according to the IoU is shown in Figure 4. This parameter configuration was used for the final model.

### Figure 4

*Calibration results*

```
=== CALIBRATION SUMMARY ===
Tried models: 25
Tried allocations: 3225

BEST OVERALL by IoU:
IoU=0.1539 TP=4167 FP=11457 FN=11457
WINDOW_EMPTY=9 WINDOW_NEIGH=3
mode=ca CA_WIN=7 CA_SEED_FRAC=0.05 CA_ALPHA=12.0
AUC=0.8204014871849276 AP=0.16743581696585177

=== FINAL (best params, full steps) ===
IoU: 0.1539 TP: 4167 FP: 11457 FN: 11457 Shown: 27081
```

### **Cellular automata**

The logistic regression model produces a probability map showing how likely each raster cell is to become housing. However, this probability map does not directly show which cells actually change. To convert these probabilities into a prediction map of new housing, a cellular automata allocation step was used.

The allocation is carried out in several steps. First, a number of cells with high transition probabilities are selected as initial “seed” locations. These seeds represent starting

points for new housing development. Next, additional cells are added step by step. Cells located near already allocated housing become more attractive, which allows housing to grow in spatial clusters rather than appearing randomly.

$$\text{score}_i = P_i \cdot (1 + \alpha \cdot n_i^{CA})$$

Where:

- $P_i$  = transition probability estimated by the logistic regression model
- $n_i^{CA}$  = number of newly allocated housing cells in the local neighbourhood
- $\alpha$  = parameter controlling the strength of neighbourhood influence

## RQ2

Moving on to the second part of our simulation modelling process, to answer the second research question which addresses our main goal in evaluating the TOD strategy proposed by D66: *“To what extent would incorporating D66’s transit-oriented development (TOD) strategy into a calibrated cellular automata-based forecasting model alter projected housing in Leidsche Rijn by 2040 compared with a baseline scenario?”*.

A scenario-based CA land use change model was implemented to project residential expansion in Leidsche Rijn from 2017 to 2040. The model builds upon the calibrated CA logistic regression framework made from RQ1 and uses identical transition drivers, constraints, and neighborhood rules to ensure comparability between scenarios.

Two scenarios were simulated: 1. Baseline scenario, representing the continuation of historical development patterns without explicit TOD strategy. 2. TOD scenario, incorporating a spatial preference for development near the two railway stations, reflecting D66’s TOD strategy.

To represent TOD, around the two railway stations within Leidsche Rijn a continuous distance raster was calculated using the Euclidean distance. An exponentially decaying function of distance was used:

$$TOD_i = e^{-di/D0}$$

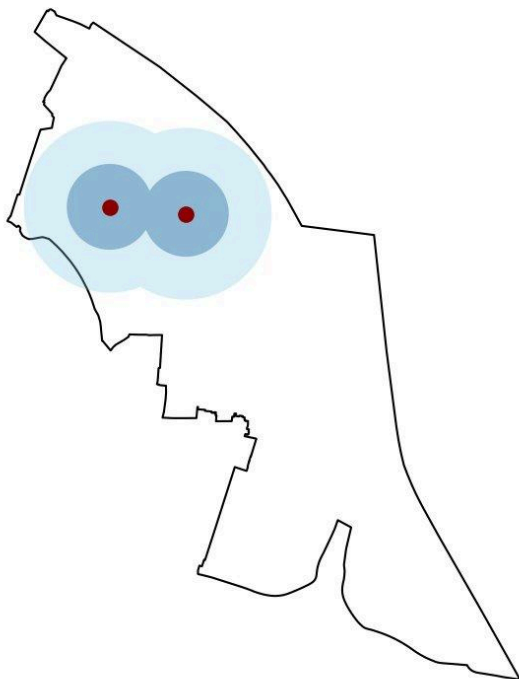
Where “ $d$ ” is the distance in meters to the nearest station and “ $D0 = 800(m)$ ” represents the walking catchment. After this function was used the following function gives the heightened TOD probability of a cell turning into residential area:

$$P_i^{TOD} = P_i^{baseline} * (1 + \beta * TOD_i)$$

The Beta ( $\beta$ ) was set at “1” after putting in multiple different values (1 to 5), since this seemed like a reasonable realistic influence according to the D66 TOD plan. The resulting TOD influence surface ranges from “2x” (at the station), “1.61x” (400m), “1.37x” (800m) and quickly lowering values further away from the station. Figure 5 shows the catchment areas of the TOD strategy around the two train stations.

**Figure 5**

*TOD stations and catchment areas (400 & 800m)*



The residential expansion was allocated using a patch-based CA algorithm, which selects candidate cells based on adjusted transition probabilities and reinforces spatial clustering through neighborhood feedback. The total residential growth between 2017 and

2040 was taken from the observed average yearly growth rate between 2010-2015 and 2015-2017. This growth was used in yearly time steps, allowing residential development near stations to progressively saturate over time.

Both the baseline and TOD scenarios were using identical growth rates and constraints, ensuring that differences in spatial outcomes are due to the TOD strategy and not because of differences in overall development size.

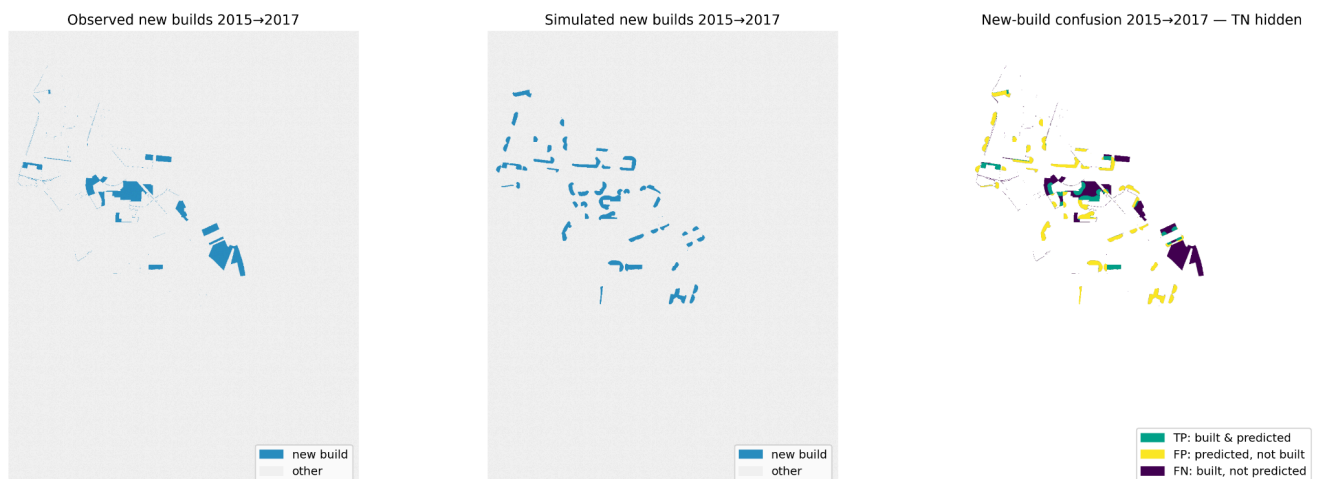
The outcomes were compared by looking at the total number of new residential cells within 400 and 800m of stations, also the final residential expansion patterns in 2040 were mapped for visual comparison between the two scenarios.

## Results

The results of the cellular automata model, to answer RQ1, are shown in figure 6. The results shown are from the calibrated model. The results show the true positive (TP); housing that was correctly predicted, false positive (FP); housing that was incorrectly predicted, and false negative (FN); housing that was not predicted. These metrics are used to calculate the inception over union ( $IOU = TP/(TP+FP+FN)$ ) which is 0.153. An enlarged version of the overlaid graphic can be seen in appendix B.

**Figure 6**

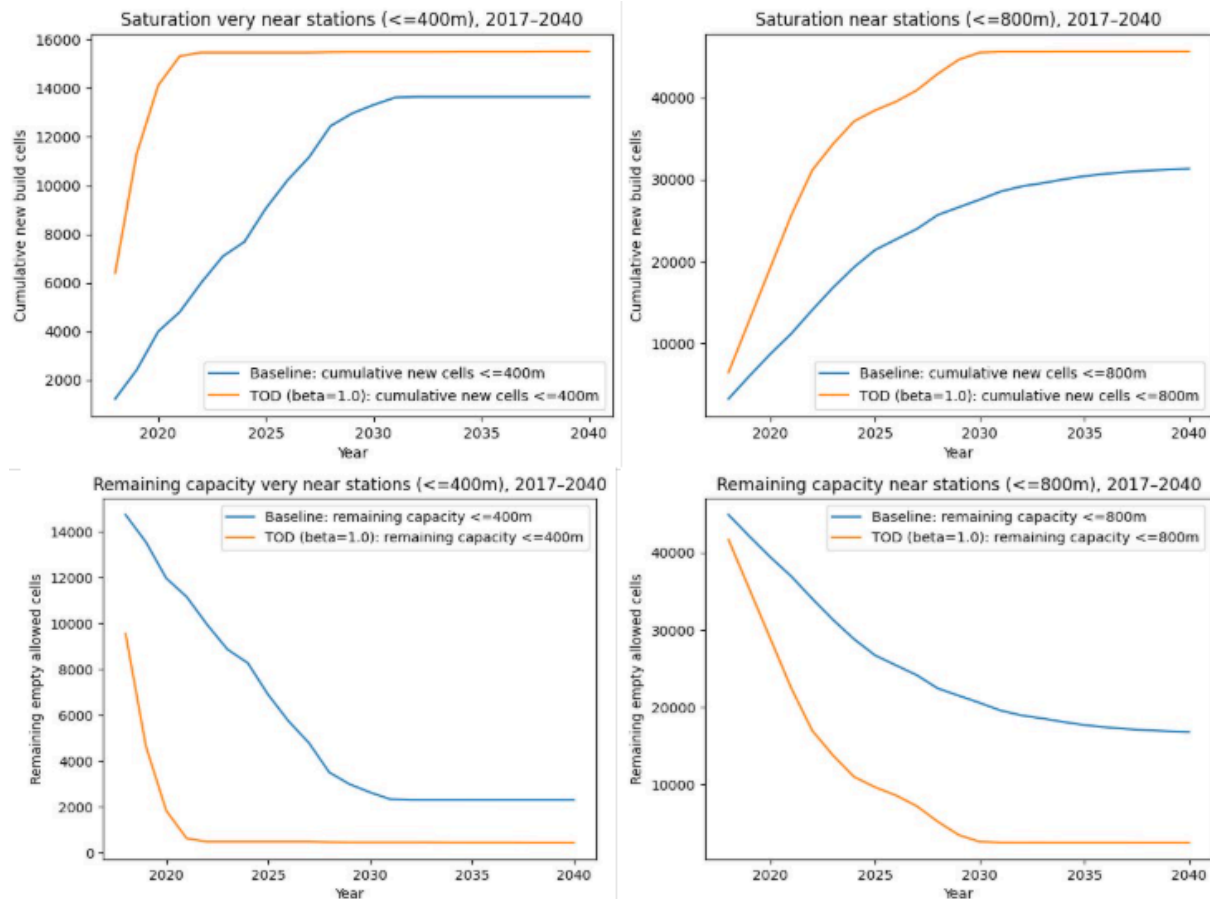
*Observed and simulated new housing (left, middle). IOU results (right)*



The results of the saturation rate and the remaining “available” space of both scenarios from the year 2017 up until 2040, to answer the RQ2, are shown in figure 7 below.

**Figure 7**

*Saturation rate and remaining capacity near stations (400 and 800m)*



The two graphs on the left show the saturation rate and the remaining capacity in the catchment, below or equal to an euclidean distance of 400m away from the train stations. The two graphs on the right show the same measures but for a distance of 800m. The saturation rate is as expected rising quicker for the 400m graph than the 800m graph due to less space available. Also does it take less time for the 400m graph to reach saturation than the 800m graph for the TOD scenario (orange line), this is due to the higher probability odds closer to the stations earlier mentioned in the methods section (400m = 2022, 800m = 2032). As can also be seen that by the year 2040 the TOD scenario has allocated substantially more new housing within 400 and 800m than the baseline scenario.

The following table shows the differences of the outcomes of the two scenarios in amount of cells and/or in percentages:

**Table 1.**

*Results table: 2017-2040, Leidsche Rijn*

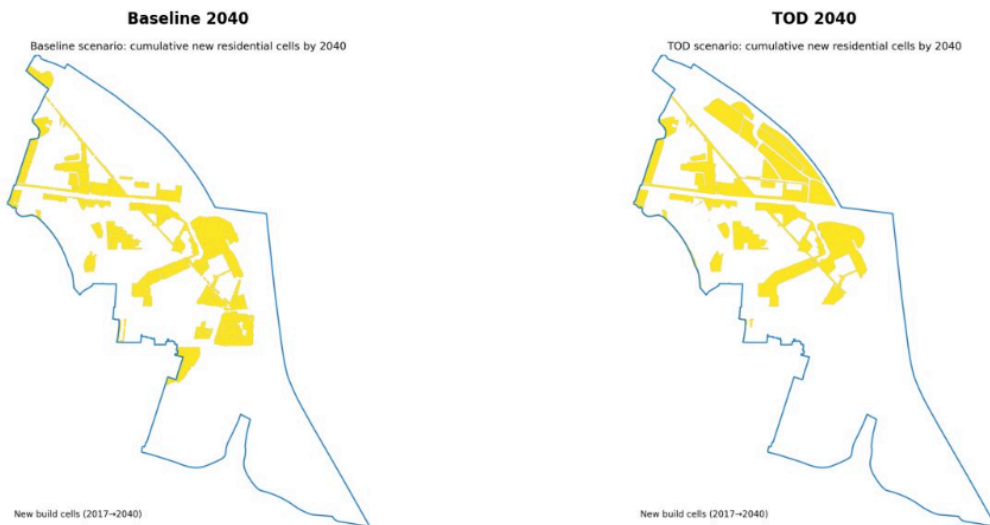
| <b>Metric (2017-2040)</b>                     | <b>Baseline model</b> | <b>TOD model (<math>\beta = 1</math>)</b> | <b>Diff (TOD - Baseline)</b> |
|-----------------------------------------------|-----------------------|-------------------------------------------|------------------------------|
| Total new residential cells                   | 147.361               | 147.361                                   | 0                            |
| New cells within 800m catchment               | 31.299 (+21,24%)      | 45.638 (+30,97%)                          | + 14.339 (+9,73%)            |
| Initial available capacity within 800m (2017) | 48.091                | 48.091                                    | 0                            |
| Percentage of 800m capacity used (2040)       | 65,08%                | 94,90%                                    | + 29.82%                     |
| Remaining capacity within 800m (2040)         | 16.792                | 2.453                                     | - 14.339                     |

Table 1 shows that the total number of new residential cells in the year 2040 is the same (both scenarios allocate 147.361 new cells). But the TOD strategy substantially changes where that growth occurs. Compared with the baseline, the TOD scenario ( $\beta = 1$ , D0 = 800m) shifts a much larger share of new residential development into the station catchments. By 2040, the number of new residential cells within 800m of the two stations increases from 31.299 (baseline) to 45.638 (TOD), which is an increase of +14.339 cells (+9.73%) of total growth (from 21.24% to 30.97%). This concentration effect also leads to near saturation of available land within 800m: The TOD scenario uses 94.40% of initially available “empty & allowed” capacity within 800m by 2040, compared with 65.08% under the baseline, leaving far less remaining capacity near stations (TOD: 2.354 vs Baseline: 16.792, available cells left).

The following figure shows the two final maps of the cumulative new residential cells that have been simulated in the period of 2017 up until 2040.

**Figure 8**

*Final maps of the cumulative new residential cells, baseline vs TOD, in 2040*



In figure 8 can be seen that the baseline model has ended up in a more widely spread simulated pattern of new cells than the TOD model. The TOD model has clustered its new cells near the train stations and has built less further away from the train stations.

In the following and final figure of this case study all the residential cells in the Leidsche Rijn are visualised. Figure 9 shows the final maps of both models in the year 2040, on the left the baseline (in green) and on the right the TOD model (in purple). Giving a final and complete picture of how far and large the residential area has expanded in 2040.

**Figure 9**

*Map of all the residential cells in the Leidsche Rijn in the year 2040, Baseline vs TOD model*





## Discussion

This study aims to evaluate the usefulness of a cellular automata–based land-use change model for analysing housing development patterns in Leidsche Rijn and for exploring the potential spatial impacts of a transit-oriented development (TOD) strategy proposed by D66. The cellular automata method was chosen after the literature review showed its wide and proven use in forecasting urban growth. The literature however also pointed to a gap in testing the effects of policy specific forecasting.

Our first research question is posited to review the effectiveness of cellular automata based forecasting in the Leidsche Rijn area. Our results suggest this method is not fit for forecasting urban growth on a local level, due to the low accuracy. However, the model implementation in this study has serious limitations which influence the effectiveness of CA-based forecasting. Firstly, the resolution of our raster was set to 5x5 meters. This may have influenced the neighbourhoods tending to become more round in shape, since the neighbourhood affects the pixels closest to the group the most. This creates a sort of “gravity effect”, forming spheroid neighbourhoods. Also, more considerations go into planning neighbourhoods than the factors our model takes into account. Our study can therefore be seen as a proof of concept within this area, providing a basis whereupon future research can improve.

The prominent difference between the baseline and TOD model suggests a good fit for planning policy evaluations. In answering our research question, implementing the TOD strategy does alter the results by a significant amount. Housing density around transit hubs has increased substantially in the TOD scenario. In this case only transit proximity has been taken into account, but this implementation can also be used for multiple factors influencing housing placement. Apart from forecasting urban growth for a homogenous population, this method could be applicable for forecasting urban growth for groups with differing priorities. Proximity to schools is for example less important for the elderly. Using multiple models to

represent differing interests could prove a useful tool for city planners in choosing where to build various types of housing.

Overall, this study has proven a cellular automata model can be used for forecasting urban growth on a local level, albeit for now on a low-accuracy level. In forecasting policy-specific urban growth, this method is excellent for visualising the tested effects. Even in the context of the D66 campaign, a projection of the Netherlands by 2040 with a (hopefully) solved housing crisis could prove as a striking marketing tool. Considering the low accuracy, the methods used in this research have room for improvement, which is detailed in the following section.

### **Recommendations for future research**

Future research could improve the robustness and interpretability of CA-based housing development models in several ways. First, training with a longer period and additional observation years would allow more reliable estimation of transition probabilities and a clearer separation between calibration and validation datasets. Similarly, exploring multiple raster resolutions could help determine the spatial scale at which housing growth patterns are most accurately captured.

In addition, future implementations of the policy scenario could strengthen the representation of TOD by increasing transition probabilities as a function of proximity to transport hubs to better reflect the spatial attraction of transit hubs and provide a more realistic simulation of TOD-driven growth. Moreover, evaluating alternative metrics beyond Intersection over Union(IoU), such as Kappa or F1-score, could offer a more nuanced understanding of model performance in forecasting housing development.

Finally, testing the model in other rapidly developing Dutch regions or under alternative TOD policy scenarios would strengthen confidence in its applicability for urban planning and policy evaluation.

## References

### Maps

CBS Land Use 2010 WMS:

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CBS Land Use 2015 WMS:

[https://service.pdok.nl/cbs/bestand-bodemgebruik-2015/wms/v1\\_0?request=getcapabilities&service=WMS](https://service.pdok.nl/cbs/bestand-bodemgebruik-2015/wms/v1_0?request=getcapabilities&service=WMS)

CBS Land Use 2017 WMS:

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## Appendix A - Collaboration Grid

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Collab Grid</b>        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Task</b>               | <ul style="list-style-type: none"> <li>- Case study proposal: preliminary version. Date <b>02/12/25</b></li> <li>- Case study: Research introduction. Date <b>19/12/25</b></li> <li>- Case study: short presentation. Date <b>13/01/26</b></li> <li>- Case study presentation. Date <b>22/01/26</b></li> <li>- Case study report. Date <b>26/01/26</b></li> </ul>                                                                                                                                                                                                                            |
| <b>Working Method</b>     | <p>We first each prepared by reading through the assignment. We then met as a group and talked about potential interesting subjects. Upon agreement on the subject we divided the work and put everything of the preliminary version in a Google Docs.</p> <p>We decided that for the rest of the case study we would meet on Tuesdays at the labs and divide the rest of the work to be done individually at home.</p> <p>Dividing the tasks is done quite naturally, at points in which people's strengths lay and based on how much people have contributed so far to the case study.</p> |
| <b>Working Atmosphere</b> | <p>Our team stays motivated in contributing to the case study by respecting each other and putting interest in what we do in our daily lives.</p> <p>The working atmosphere feels quite healthy. There is lots of room for input and alterations in our course.</p>                                                                                                                                                                                                                                                                                                                          |
| <b>Coaching</b>           | <p>Everyone works on a different part of the case study and when someone comes to a stop he asks for some feedback or help.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

**Appendix B - Excerpt from Figure 6**

New-build confusion 2015→2017 — TN hidden

